A MORPHISM IS DETERMINED ON A NEIGHBOURHOOD BY ITS MAP ON LOCAL RINGS, AND A LOCAL MAP OF LOCAL RINGS IS REALIZED BY A MORPHISM, IN FOUR STANDARD SITUATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-morphism-defined-local-ring is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 10338-10366 and proof span 10368-10422 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The retained author text contains a typographical slip in its last sentence ('by another application of See Algebra, Lemma ...'). Under the lane rule that all delivered mathematical text is copied from the pinned chapter file without rewriting, the sentence is retained verbatim and no correction is made.

Editorial note. Tag self-check per LANE_BRIEF 10.5 (5): the label lemma-morphism-defined-local-ring resolves in the project's own tag list to canonical tag 0BX6 (line 14969 of the index, full label morphisms-lemma-morphism-defined-local-ring). reference/public-items.jsonl holds no row whose primary source_file is morphisms.tex, so the intersection with the same-chapter public tag ids is empty and the candidate passes.

Editorial note. The gate was run as: python operations/agent-authoring/tools/precard_gate.py –source morphisms.tex –label lemma-morphism-defined-local-ring –start 10338 –end 10366 –proof-start 10368 –proof-end 10422 –local-id 34091, giving RESULT=OK with all four span guards passing, no public interval row for this chapter and occupancy FREE for morphisms.tex 10338 10422. Spans were recomputed live from the label in the pinned chapter file; the located text matched the pool byte counts (statement 1038 characters, proof 2152 characters after whitespace removal). The proof span contains a commutative diagram drawn with xymatrix, which the retained 88-line chapter preamble supports.

Editorial note. The candidate was newly offered by the queue only after the handed-over windows were filtered out of the queue helper, so the counting region of this card is outside the windows 8000-10000 and 12000-14000 of this chapter that were transferred to lane-s1.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A subring of a local ring contains the localization of the subring at an element outside the maximal ideal, cited from the algebra chapter. Cited at proof lines 10388-10389 and 10419-10420. Produces the element g outside the relevant prime over which the two maps agree, and later the element over which a map from the base into the local ring factors.
- The characterization of finite presentation, and the description of a localization as a filtered colimit, cited from the algebra chapter. Cited at proof lines 10410-10412. Lets the given local map from a finitely presented algebra be realized, after localizing the source at one element, by an honest algebra map, which is what produces the required morphism in the first case of part (2).

Source reference. Original lemma 00QO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 974–974. Same-chapter reference used in the retained author text at source line(s) 10408, 11936. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 00CR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 471–471. Same-chapter reference used in the retained author text at source line(s) 10409. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0BX1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14964–14964. Same-chapter reference used in the retained author text at source line(s) 10389, 10421. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let S be a scheme. Let X, Y be schemes over S. Let $s \in S$ and $x \in X$, $y \in Y$ points over s.*

(1) *Let $f, g : X \to Y$ be morphisms over S such that $f(x) = g(x) = y$ and $f_x^\sharp = g_x^\sharp : \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$. Then there is an open neighbourhood $U \subset X$ with $f|_U = g|_U$ in the following cases*
   (a) *Y is locally of finite type over S,*
   (b) *X is integral,*
   (c) *X is locally Noetherian, or*
   (d) *X is reduced with finitely many irreducible components.*

(2) *Let $\varphi : \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ be a local $\mathcal{O}_{S,s}$-algebra map. Then there exists an open neighbourhood $U \subset X$ of x and a morphism $f : U \to Y$ mapping x to y with $f_x^\sharp = \varphi$ in the following cases*
   (a) *Y is locally of finite presentation over S,*
   (b) *Y is locally of finite type and X is integral,*
   (c) *Y is locally of finite type and X is locally Noetherian, or*
   (d) *Y is locally of finite type and X is reduced with finitely many irreducible components.*

Proof. Proof of (1). We may replace X, Y, S by suitable affine open neighbourhoods of x, y, s and reduce to the following algebra problem: given a ring R, two R-algebra maps $\varphi, \psi : B \to A$ such that

(1) $R \to B$ is of finite type, or A is a domain, or A is Noetherian, or A is reduced and has finitely many minimal primes,
(2) the two maps $B \to A_{\mathfrak{p}}$ are the same for some prime $\mathfrak{p} \subset A$,

show that φ, ψ define the same map $B \to A_g$ for a suitable $g \in A$, $g \notin \mathfrak{p}$. If $R \to B$ is of finite type, let $t_1, \ldots, t_m \in B$ be generators of B as an R-algebra. For each j we can find $g_j \in A$, $g_j \notin \mathfrak{p}$ such that $\varphi(t_j)$ and $\psi(t_j)$ have the same image in A_{g_j}. Then we set $g = \prod g_j$. In the other cases (if A is a domain, Noetherian, or reduced with finitely many minimal primes), we can find a $g \in A$, $g \notin \mathfrak{p}$ such that $A_g \subset A_{\mathfrak{p}}$. See Algebra, Lemma 0BX1. Thus the maps $B \to A_g$ are equal as desired.

Proof of (2). To do this we may replace X, Y, and S by suitable affine opens. Say $X = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(B)$, and $S = \operatorname{Spec}(R)$. Let $\mathfrak{p} \subset A$ be the prime ideal corresponding to x. Let $\mathfrak{q} \subset B$ be the prime corresponding to y. Then φ is a local R-algebra map $\varphi : B_{\mathfrak{q}} \to A_{\mathfrak{p}}$. If $R \to B$ is a ring map of finite presentation, then there exists a $g \in A \setminus \mathfrak{p}$ and an R-algebra map $B \to A_g$ such that

$$\begin{array}{ccc} B_{\mathfrak{q}} & \xrightarrow{\varphi} & A_{\mathfrak{p}} \\ \uparrow & & \uparrow \\ B & \longrightarrow & A_g \end{array}$$

commutes, see Algebra, Lemmas 00QO and 00CR. The induced morphism $\operatorname{Spec}(A_g) \to \operatorname{Spec}(B)$ works. If B is of finite type over R, let $t_1, \ldots, t_m \in B$ be generators of B as an R-algebra. Then we can choose $g_j \in A$, $g_j \notin \mathfrak{p}$ such that $\varphi(t_j) \in \operatorname{Im}(A_{g_j} \to A_{\mathfrak{p}})$. Thus after replacing A by $A[1/\prod g_j]$ we may assume that B maps into the image of $A \to A_{\mathfrak{p}}$. If we can find a $g \in A$, $g \notin \mathfrak{p}$ such that $A_g \to A_{\mathfrak{p}}$ is injective, then we'll get the desired R-algebra map $B \to A_g$. Thus the proof is finished by another application of See Algebra, Lemma 0BX1. $\square$